

Multimedia Databases Exercises

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Exercise 5

Subject: Images Processing Part 2

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Task 1: Point Operations

In this exercise, we assume that we are using an 8-Bits grayscale image. The *HK* point operation is defined as follows:

$$P_{output} = \alpha P_{input} + \beta$$

P_{input} and P_{output} : Pixel values of the input and output image respectively.

1. How do the parameters α and β influence the result of the operation?
2. Explain which HK operation could be applied to implement the image inversion function.
3. Which problems could appear, if *HK* is used with unadapted α and β parameters? Propose a method to deal with these effects.
4. Let G be a grayscale image with minimum pixel value a and maximum pixel value b . Which HK Operation could be applied to G in order to maximize its contrast ratio?

Task 2: Linear filter

Figure 1 describes the region processing operation called *convolution*. The application of the operation on all pixels of an image defines a linear filter.

1. Which problems can occur to edge pixels when using this filter? Propose possible approaches to deal with these problems.
2. A smoothing filter can be implemented using a linear (convolution) filter and a well-chosen kernel. For this purpose calculate a 5x5 kernel for the 'moving average' variant of the smoothing filter, in which a pixel is replaced by the average values of its neighbours.
3. Weighted smoothing: in this case pixels which are closer to the target pixel get higher coefficients. The coefficients can be calculated using 2D-functions, see for example:

Linear Filter

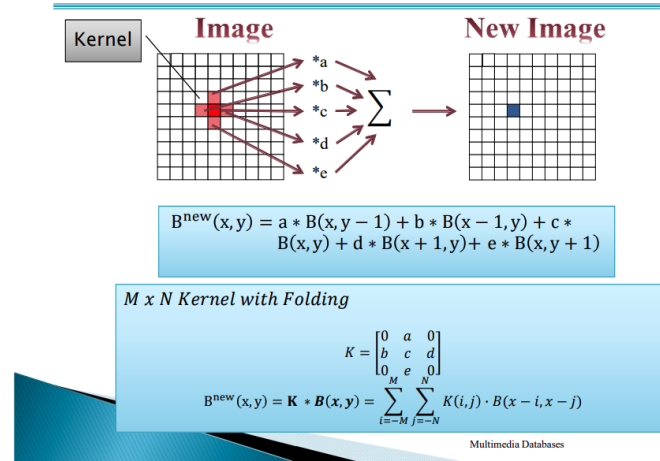


Figure 1: Convolution

- the pyramid-area-function:

$$f(x, y) = -\alpha \cdot \max(|x|, |y|) + k$$

x and y designate the distance to the target pixel on the x and y axes, α is a parameter of the function and k a constant which you add for creating positive values.

- the conical area function:

$$f(x, y) = -\alpha \cdot \sqrt{x^2 + y^2} + k$$

Calculate the coefficients of a 5x5 kernel for these 2 cases with $\alpha = 2$. Select an adequate k so that the smallest coefficient is equivalent to zero, and round the values to get integers.

- Why are laplacian filters used? Give an example.